CHENG-YAN TSAI

GAME PROGRAMMER | TECHNICAL ARTIST | GRAPHICS PROGRAMMER

College Station, TX | +1 979-574-8872 | crescentmax@tamu.edu | crescentmax.github.io

SUMMARY

Game programmer and technical artist with Unreal Engine experience in AAA gameplay, VR, and real-time graphics. Builds scalable gameplay systems, tools, and content pipelines for multidisciplinary teams.

PROFESSIONAL EXPERIENCE

Unreal Developer — Graduate Research Assistant **Feb 2025 – Present**

Soft Interaction Lab, Texas A&M University

CineDesk — Virtual Production Research

- First-authored SIGGRAPH-accepted research on a touch-input interface for virtual production workflows.
- Evaluated a tangible SpaceMouse peripheral to improve camera-control precision and user interaction.
- Implemented 12 production features, including camera translation, lens adjustment, preview, and staging controls.

Forensic Nurse Training

- Developed an interactive Unreal Engine mock trial with a custom LLM server in a 5-person team.
- Engineered real-time lip sync that loads TTS audio and maps visemes to MetaHuman characters.
- Partnered with the Texas A&M School of Nursing to align AI responses with medical and legal standards.

Unreal Game Engineer **Aug 2023 – Jul 2024**

Winking Studios

Project Kyzen — Shipped

- Delivered gameplay features for a commercial metaverse title across two on-schedule production releases.
- Collaborated with designers, artists, and engineers across a 40-person international team.

AAA Action RPG — Unannounced, Early Stage

- Designed and implemented three core gameplay systems within a 30-person development team.
- Created a scalable dialogue pipeline that accelerated narrative iteration by 35%.
- Built a reusable interaction framework adopted by three teams to standardize gameplay interactions.

Unreal Developer — Undergraduate Research Assistant **Jul 2021 – Dec 2022**

CGV/MIS Lab, National Tsing Hua University

- Built a real-time ray-tracing renderer for AI virtual staging driven by JSON scene configurations.
- Improved rendering throughput by more than 60% through a high-speed Unreal-based rendering workflow.
- Generated 1,000+ synthetic images that improved dual-hand humanoid-robot manipulation accuracy by 20%.

SELECTED PROJECTS

Digital Twin for Retail Stores | *Unreal Engine* — Built a framework for simulating store operations and customer experiences to support retail optimization.

Physically Based Simulations | *Unreal Engine / OpenGL* — Implemented real-time rigid-body, soft-body, fluid, and flocking simulation systems.

Lucid Dive | *Unity* — Created a 2D action platformer with 1,800+ itch.io views; received Best Graphics and Best Music Design.

Orbitone | *Unity VR* — Developed a VR rhythm game that placed 2nd among 15 teams in a VR competition.

TECHNICAL SKILLS

Programming: C/C++, Python, JavaScript, Ruby, HTML

Engines & Graphics: Unreal Engine, Unity, Godot, OpenGL, Real-Time Rendering, Ray Tracing, Physics Simulation, VR

Technical Art & Tools: Blender, Houdini, MotionBuilder, Substance Designer

EDUCATION

Master of Computer Science **Aug 2024 – May 2026**

Texas A&M University • College Station, TX

B.S., Electrical Engineering & Computer Science **Sep 2020 – Jun 2023**

National Tsing Hua University • Taiwan

B.S. in Computer Science — Transferred **Sep 2018 – Jun 2020**

Huazhong University of Science and Technology • China